

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Portable Double-Focus Mass Spectrograph with Multymembrane Inlet

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ABSTRACT

Portable Mass Spectrometer is designed for direct permanent autonomous in situ analysis of mixture's composition in real time scale. Small size and weight, low consumption of the instrument and high sensitivity to organic compounds allow to utilize it for environmental investigation or technological control outside the laboratory. Simple design and small permanent magnets that require no power at all make static Mass Spectrometer attractive for miniaturization. Mass Spectrography Mode gives the best results due to possibility of monitoring of several compounds simultaneously. Software package of numerical simulation allowed to calculate the ion-optic scheme of the original Double-Focus Mass Spectrograph with curved magnet outline. This scheme supplies high dispersion and small sector angle of the magnet for the heavier masses. A number of volatile and semivolatile organic compounds have a high permeability through the silicone membrane in comparison with the nature matrix (air, water) permeability. Based on this property the multymembrane inlet was created. It allows to increase the sensitivity of Mass Spectrometer for organic impurities in air million times more. Multimembrane inlet system involving consecutively disposed membranes ($N > 2$) enriches sample with compounds effectively in MS selective analysis. Double-Focus Mass Spectrograph with Multymembrane Inlet is a real prototype of the portable instrument.